National University of Singapore
School of Computing

Learning to Coordinate through Strategic Game Slices

Final Year Project
Continuous Assessment Report

Student	Li Daoxin
Student number	A0288040H
Project code	H009720
Supervisor	Chua Tat Seng
Main evaluator	Yang YOU
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Abstract

Coordination under private information requires agents to infer others' preferences and use those inferences to choose effective actions. This project studies these capabilities through strategic game slices: selected segments of a finite negotiation game with binding commitments and private investigation. A reference-policy solver provides preference posteriors, action values, and information-omission losses. Matched diagnostic interventions reveal that supplying a correct preference judgment does not reliably resolve planning errors. Initial training experiments show improved calendar coordination, with joint slice training retaining 47 of 72 meetings compared with 37 for the base model and complete-game self-play. Further analysis identifies two obstacles to effective supervision: information-sensitive decisions are sparse, and grouped sampling often produces identical actions even at persistently incorrect decisions. Controlled game-design studies examine how hidden preferences and commitment structure affect this signal. The next stage will use these findings to construct a targeted slice curriculum and evaluate its effect on negotiation and transfer.

1 Introduction

Large language models (LLMs) increasingly participate in multi-agent tasks that require communication, negotiation, and coordinated action. In these settings, an agent's success depends on information distributed across participants and on the decisions those participants make. Interactive benchmarks capture this dependence through collaboration, competition, and decentralised scheduling [5, 11]. However, a failed interaction rarely reveals whether an agent misunderstood its partner or selected an ineffective action despite sufficient understanding. This ambiguity makes both diagnosis and training difficult.

We study two coupled capabilities: *interactive-state inference*, which estimates a partner's task-relevant hidden preferences from available evidence, and *interaction planning*, which uses this information to evaluate and select actions. The distinction follows classical accounts of decision-making with models of other agents [1, 3]. It also suggests a practical intervention: compare an agent's choices when supplied with its own preference judgment and with a correct oracle judgment. The remaining error measures a planning difficulty that preference prediction alone does not capture.

Our approach uses a finite negotiation game in which players make binding commitments toward goals with aligned or conflicting preferences. Private preferences create uncertainty, while investigation provides an opportunity to acquire information. A reference solver evaluates the consequences of legal decisions, allowing the same environment to support diagnosis and training. We use *game slices*, interaction segments beginning at selected decision contexts, to concentrate supervision on identifiable inference and planning demands. This gives finer control over training content than relying entirely on states visited in complete-game self-play.

The project investigates three questions. First, how much action error remains after a correct preference judgment is supplied? Second, which game structures create information

that changes an optimal decision? Third, can training on selected decisions improve complete-game behaviour and coordination in a different environment? These questions connect the design of a training task to the reasoning difficulty it presents and the learning signal it actually produces.

The contributions completed at this stage are as follows:

1. We develop a negotiation environment and reference-policy evaluation framework that supports matched tests of preference inference, planning, and the value of observed information.
2. We conduct diagnostic and training experiments with Qwen3-4B-Instruct-2507, identifying persistent planning errors under oracle judgments and preliminary transfer gains from slice training.
3. We characterise sparse supervision through training-log analysis and controlled three-player game studies, showing how commitment choices and goal dependencies affect information-sensitive decisions.

The report presents the framework, completed experiments, and the research plan for a revised slice-training study. The new game-design candidates have completed solver-based analysis; model evaluation and training on these candidates form the next phase.

2 Literature Review

Inference and planning. Interactive partially observable Markov decision processes represent uncertainty over both the environment and other agents, connecting belief updates to action selection [1]. Bayesian Action Decoder further shows how an action's informational meaning depends on the policy that produced it [2]. Hierarchical Opponent Modeling and Planning couples goal inference with response planning in mixed-motive environments [3]. For language agents, theory-of-mind representations have also been studied as a mechanism for collaboration [4]. These approaches motivate our separation of preference inference and conditional planning. We use an explicit reference policy to interpret behaviour and matched interventions to test whether an inferred preference is used effectively.

Coordination benchmarks. MultiAgentBench evaluates collaboration and competition across interactive scenarios [5]. Benac et al. model negotiation through binding action-level commitments, exposing the importance of delayed consequences [6]. CalBench studies communication across private calendars and evaluates scheduling quality against reference solutions [11]. Our negotiation environment extends the local commitment-game implementation with private preferences and investigation. Its finite structure permits decision-level labels, while the project's sequential CalBench adaptation tests transfer to a different action space.

Game-based learning and curricula. MARSHAL trains strategic LLMs through multi-turn self-play, using turn-level credit assignment and agent-specific advantage normalisation [7]. Subgame curriculum learning instead adjusts the distribution of starting states to focus practice on useful parts of a game [8]. These approaches motivate our use of selected interaction segments. We optimise language-model responses using grouped relative advantages, following the principle of GRPO [9], and examine whether the sampled responses provide informative reward differences. Other-Play's analysis of partner-specific conventions also motivates evaluation beyond homogeneous self-play [10].

This project connects these directions by using one controlled environment for capability diagnosis, decision-level supervision, and curriculum construction. The central issue is whether a selected context combines consequential information, a model error, and a usable optimisation signal. The experiments examine these properties separately.

3 Strategic Game Slices

3.1 Negotiation under Private Information

The game contains players $i \in \{0, \ldots, n-1\}$, binary commitment variables x_j, and goals g. Each commitment belongs to one player. A goal specifies a set of required commitments R_g and uses binary or linear completion scoring:

$$f_g(x) = \prod_{j \in R_g} x_j \quad \text{or} \quad f_g(x) = \frac{1}{|R_g|} \sum_{j \in R_g} x_j. \tag{1}$$

Player i has preferences $\theta_{ig} \in \{-1, 0, 1\}$, corresponding to avoid, neutral, and want. Its terminal utility is

$$U_i(x, \theta_i) = \sum_g \theta_{ig} f_g(x). \tag{2}$$

The same goal can therefore benefit one player and harm another. Players negotiate over contributions to these shared goal structures.

At a proposal opportunity, a player can make a bilateral offer, pass, or investigate an eligible hidden preference. An offer specifies additional commitments for the proposer and recipient. Acceptance makes them binding; rejection leaves the commitment state unchanged. Commitments accumulate and cannot be withdrawn. Proposal opportunities follow a fixed round-robin schedule, and the recipient's response does not consume its own future proposal opportunity.

Each player observes public preferences, its own complete preferences, and the public action history. An investigation reveals one true preference privately to the investigator, while the queried player and goal are publicly observable. Subsequent decision inputs include the player's accumulated answers. The recent research candidates use independent uniform priors over hidden preferences and exclude queries of publicly disclosed or previously directly queried preferences. Behavioural inference does not remove a query target from the legal menu.

The game-design studies also introduce a personal offer quota. Proposing an offer consumes this quota immediately, including when the recipient rejects it. Investigation and passing consume proposal opportunities. The resulting cost is the loss of a future opportunity rather than a utility penalty. The three-player studies use two round robins, one investigation and one offer per player, with additional commitment choices introduced in controlled variants. These parameters define the studied instances within a more general game framework [24].

3.2 Reference Policies and Action Values

A behavioural reference policy specifies a distribution over legal actions at each information context. Strategies respect the player's observations: worlds that are indistinguishable to a player share the same decision rule. Candidate references are checked numerically for unilateral improvement, locally optimal supported actions, and the response objective. At ACCEPT/REJECT decisions, players maximise their own expected terminal utility and, in an own-utility tie, maximise the sum of other players' expected terminal utilities. Proposal decisions use own utility alone. The numerical checks use equilibrium tolerance 10^{-8} and tie tolerance 10^{-9}; saved policies are independently reloaded and rechecked [19].

For a focal player i at information context I, let π_{-i}^{ref} denote the fixed policies of its opponents. The action value and regret are

$$Q_i(I, a) = \max_{\pi_i:\, \pi_i(I)=a} \mathbb{E}[U_i \mid I, \pi_i, \pi_{-i}^{\mathrm{ref}}], \tag{3}$$

$$R_i(I, a) = \max_{b \in \mathcal{L}(I)} Q_i(I, b) - Q_i(I, a), \tag{4}$$

where $\mathcal{L}(I)$ is the legal action set. The focal player's future contingent strategy is optimised to the specified endpoint. In the recent main analysis, that endpoint is the native terminal state. Thus Q captures the consequences of later offers and responses, rather than only the immediate change in commitments.

3.3 Information Sensitivity

We measure the decision value of an observed item e by comparing a full-information optimal policy π_I^* with a policy π_{-e}^* that cannot access the item:

$$S_e(I) = V_I(\pi_I^*) - V_I(\pi_{-e}^*). \tag{5}$$

The omitted item is either one private investigation answer or one completed public proposal event. Entry contexts with identical retained observations share a policy under omission. Both policies are evaluated at the same true context, with the same opponents and terminal utility. Rules, current commitments, remaining quotas, and other observations are retained, and later evidence may reveal the missing information again.

The omission policy is optimised over a forest of indistinguishable entries. Their weights

use counterfactual prefix likelihood, excluding the focal player's past action probabilities. This prevents its earlier choices from revealing the hidden item. After own optimisation and the response tie rule, remaining omission-policy ties are mixed uniformly. Entries with zero counterfactual likelihood have undefined S and are excluded from positive/zero statistics. Selected witnesses receive independent sequence-form linear-programming checks, including bounds over own-optimal omission policies [19].

We also record the regret of the omission policy's first action. This distinguishes an immediate contract-selection error from a loss that arises only in later decisions. Investigation value is measured separately at the pre-query context by comparing its Q with the best non-query action. It averages over subsequent outcomes, whereas $S_e(I)$ measures the use of information already available at a particular context.

3.4 Slice Supervision

A game slice begins at a selected information context and specifies a focal player, a model-control horizon, and a continuation policy. Historical experiments explore three objectives: direct action selection, explicit preference inference, and planning conditioned on a supplied oracle judgment [13]. These objectives train a shared model; they do not require an explicit inference-to-planning pipeline at evaluation.

For direct action selection, the terminal-slice implementation uses negative own-utility regret as the task reward. Multiple responses are sampled at the same decision context and compared within their group. Training therefore depends on both the structure of the task and the actions the model samples. If all responses receive the same reward, there is no within-group task-relative signal even when an alternative legal action has higher value.

Curriculum construction additionally requires a model-difficulty measurement D: empirical performance under a specified checkpoint, prompt, tools, decoding settings, and continuation. Q identifies valuable actions, S identifies consequential information, and D identifies the model's actual difficulty. The recent game-design studies establish Q/S; their D measurement and revised slice training remain future work.

4 Model Diagnosis and Preliminary Training Results

4.1 Preference Inference and Conditional Planning

We evaluate Qwen3-4B-Instruct-2507 [12] on 22 two-player diagnostic cases. Each case elicits a preference judgment, a direct action, a planning action conditioned on the model judgment, and a planning action conditioned on the oracle judgment. The planning calls remove behavioural history while retaining the state and a judgment sufficient to determine the optimal action set. This intervention tests the decision made from a supplied inference [14, 13].

Table 1 shows failures in both behavioural inference and conditional planning. The model recovers every directly revealed preference but correctly plans in only three of

Table 1: Base-model diagnostic accuracy. Joint inference requires both posterior support and the most likely preference to match the oracle.

Evidence	Inference	Direct	Plan: model	Plan: oracle
Direct investigation answer	6/6	2/6	3/6	3/6
Single-choice elimination	0/6	3/6	2/6	3/6
Ambiguous single choice	5/6	2/6	2/6	2/6
Multi-step update	0/4	4/4	4/4	3/4
Total	11/22	11/22	11/22	11/22

Table 2: September checkpoint results on 24 sequential scheduling games. Checkpoint identifiers connect the training families to their archived runs.

Training family (checkpoint)	Meetings /72	Full games /24
Base model (Q0)	37	5
Complete-game self-play (SP-41)	37	6
Direct action (O-99)	38	8
Action and planning (BP-99)	44	10
Joint objectives (micro24-39)	47	9

these six cases even with the oracle judgment. Equal aggregate action accuracy also conceals changes in utility: among ten valid pairs with initially incorrect judgments, oracle replacement improves the selected action in three, leaves utility unchanged in six, and reduces it in one. This motivates supervising action selection alongside preference judgments.

4.2 Transfer to Sequential Calendar Coordination

The preliminary transfer experiment uses a local sequential adaptation of CalBench [11]. Four agents coordinate three meetings in each of 24 fixed scenarios. Agents observe private calendars and communicate to establish common meeting slots while preserving earlier commitments. Every player in a team uses the same checkpoint. We report meetings retained in the final calendars and games retaining all three meetings.

Joint training retains ten more meetings than the baseline and self-play, while action-and-planning training completes the most full games (Table 2). The results suggest that conditional planning is a useful training component for transfer. These are preliminary checkpoint comparisons: training exposure and serving protocols vary across the archived runs [15]. A controlled comparison with matched budgets and repeated runs is planned for the next phase.

4.3 Terminal-Slice Learning Dynamics

A subsequent terminal-slice run used the same base model on one H100-96, with eight responses per node group and continuous negative-regret rewards. It completed 307

Table 3: Validation during terminal-slice training (job 960070).

Updates	Whole-slice success /96	Entry accuracy	Mean regret
0	49	64.58%	0.19520
119	72	79.17%	0.08249
270	80	84.38%	0.04725

updates and 1,123,499 generated training tokens before cancellation. The last saved checkpoint corresponds to 270 updates. This run used JSON observations and an indexed `act` tool; later versions restored readable observations and named action tools. Results here refer to the indexed-interface run [16, 17].

Validation improves substantially (Table 3), but training signal becomes sparse. Only 140 of 6,279 node groups have nonzero within-group task-relative advantage. During the final 57 updates, 98.72% of groups sample a single distinct legal action, despite having multiple legal alternatives. Consequently, some consistently incorrect entries produce the same negative reward across all eight responses. The twelve training entries requiring investigation receive no correct response in 3,272 samples after update 50. The validation set contains no information-positive investigation entry and therefore misses this failure [16].

These observations distinguish task difficulty from learnability under the current sampling distribution. More informative rewards help when sampled actions have different values; they cannot create a relative signal among identical choices. This motivates measuring sampling diversity and investigation performance explicitly in the revised experiment.

5 Game Structure and Information-Sensitive Decisions

5.1 Varying Hidden Preferences and Commitment Options

The October reference-policy study obtains terminal-certified labels for 99 of 100 parent games. A separate finite-window evaluator solves all 100, but provides different continuation values. In the terminal set, only two of the 33 measurable three-player games contain positive own information-omission loss [19]. These games initially have one hidden preference and one commitment per player. We investigate whether richer information and contract choices create additional decision-relevant uncertainty.

The H2 variant adds a second hidden preference belonging to another player, while retaining two round robins, one investigation, and one offer per player. Of 33 attempted games, 31 receive terminal certification and Q/S labels. Nine contain positive own S, comprising 49 of 183,348 measurable contexts [20]. The A2H2 variant adds a second commitment option to one player, with both options available for cumulative agreement. Thirty of 33 games receive labels. Table 4 compares the 29 parents measured in both arms.

Table 4: Paired three-player results on 29 jointly measurable parents. H2 has two hidden preferences; A2H2 additionally gives one player a second commitment option.

Quantity	H2	A2H2
Parents with positive own S	8	9
Positive contexts	43	632
Measurable contexts	172,033	414,853
Positive contexts with current-action regret	39	585
Positive contexts on reference paths	6	193
Positive share of reference decision occupancy	0.0244%	0.1080%

Table 5: Structure-coverage results for the 22 certified parents.

Quantity	Result
Attempted / certified and measured parents	24 / 22
Parents with positive own S	14
Private-answer / public-history positive contexts	87 / 27
Positive / measurable contexts	114 / 246,713
Positive contexts with current-action regret	112
Positive share of reference decision occupancy	0.5476%

Additional commitment choices create more information-sensitive contexts, but the gain is concentrated: two parents account for 94.8% of A2H2's positive contexts [21]. This concentration makes parent-level and structural coverage important when constructing a curriculum. It also suggests examining the goal dependencies that connect commitments, rather than increasing the number of rounds alone.

5.2 Structure-Coverage Pilot

We generate an independent set of 24 parents organised into 12 pairs. The design covers split and bundled goal structures, binary/mixed/linear scoring, and alternative hidden-preference placements. A bundled goal requires both commitment options belonging to the same player; split structures avoid this dependency. Each pair compares uniform hidden placement with placement on two contributors to a common goal. Hidden preferences remain independent and uniform, and candidates are generated without resampling based on Q/S [23].

All 24 games are attempted, and 22 obtain terminal-certified labels and pass independent reload checks. Indices 4 and 11 remain unlabelled after retries. Successful cases reach the native terminal state with no nonterminal cutoff leaves. Fourteen parents contain positive own S, although the positive-context share remains 0.0462% (Table 5). Two parents contribute 75 of the 114 positive contexts. The reference occupancy measure weights decisions by the saved policy's visitation probabilities; it describes oracle coverage rather than the model's observed difficulty.

Among measured parents, bundled structures contain positive own S in 9/11 games, compared with 5/11 split games. Hidden placement shows a channel-dependent pattern.

Table 6: Goal requirements and preferences in the case study. The realised hidden values are shown for analysis; each is initially private to its owner.

Goal	Required commitments	P0	P1	P2
G0	B, C, D	Neutral	Neutral	Avoid
G1	A, B, C	Want	Neutral	Neutral
G2	C, D	Want (hidden)	Avoid	Want (hidden)

Table 7: P0's action values at its second proposal, including the final P1 proposal and responses under the saved reference.

Action	Q
Pass	2/9
Offer to P1: add B only	1/2
Offer to P1: add C only	2/9
Offer to P1: add A only	5/12
Offer to P1: add A, B	2/3
Offer to P1: add A, C	2/9
Offer to P2: add D only	23/30
Offer to P2: add A only	5/12
Offer to P2: add A, D	17/18

Across ten complete pairs, uniform placement gives private-answer-positive parents in 5/10 cases and shared-goal placement in 4/10; including history sensitivity changes the counts to 5/10 and 7/10. These results motivate goal dependency as a coverage dimension while retaining both placement strategies. The sample is small, and the two missing games both use linear scoring.

5.3 Case Study: Information Changes the Preferred Contract

A three-player game from the structure pilot illustrates the underlying decision [24]. P0 controls commitment A, P1 controls B, C, and P2 controls D. All goals use linear scoring (Table 6). Two hidden preferences, P0/G2 and P2/G2, are independent and uniformly distributed, giving nine possible worlds. We consider the realised world in which both are want.

The proposal order is P2, P0, P1, P2, P0, P1. P2 first offers to obtain A without adding a commitment, and P0 rejects. P0 investigates P2/G2 and learns want. P1 then investigates P0/G2, and P2 passes. At P0's second proposal, all commitments are inactive. P0 retains its own investigation answer, and P2's offer quota has been consumed by its first proposal. Table 7 lists all actions at this context.

With the answer available, P0 optimally offers $A + D$ to P2, which accepts. P1's final reference proposal mixes equally among three own-utility-tied choices. These produce P0 payoffs 5/6, 7/6, and 5/6, giving

$$Q_i(I, A + D) = \frac{1}{3}\left(\frac{5}{6} + \frac{7}{6} + \frac{5}{6}\right) = \frac{17}{18}. \tag{6}$$

This value includes future negotiation and depends on the specified reference mixture at P1's proposal.

When the private answer is omitted, P0 must use one policy across the three answer contexts. The omission-optimal action offers $A + B$ to P1, yielding 2/3 in the realised want context. Therefore

$$S_{\text{answer}}(I) = \frac{17}{18} - \frac{2}{3} = \frac{5}{18}. \tag{7}$$

The neutral and avoid contexts retain the same optimal $A+B$ offer and have zero omission loss. Information thus changes which partner and contract P0 should choose. Replaying 448 native continuation nodes matches the three context-specific Q/S records, and an independent LP gives the loss interval $[5/18, 5/18]$ over own-optimal omission policies for this fixed reference. At the earlier pre-query decision, the investigation's advantage over the best non-query action is 71/810. The difference between these quantities reflects acquisition before observing the answer and utilisation at the realised later context.

6 Discussion

The results identify three requirements for a useful slice curriculum. A game must offer strategically different actions, observations must help select among them, and the model's samples must expose a reward contrast. These requirements can fail independently. More hidden preferences create uncertainty, but many answer groups still share a common optimal action. Additional commitments create new contracts, yet positive S may remain concentrated in a few parents. Finally, a meaningful Q difference provides no within-group task signal when all responses choose the same action.

The response objective also affects what counts as relevant information. An H2 analysis finds 395 response contexts sensitive to the social tie rule despite zero main own S; 277 terminal responses have verified losses to other players [22]. The focal scorer currently uses own regret, while the reference distinguishes these ties. One recorded response has equal own values but other-player totals of 2 and 4, with both actions receiving the same task reward [18]. Aligning response evaluation and reward is therefore part of the next methodological step.

The present evidence is conditional on the saved reference policies and numerical certification. Independent LP checks establish selected witnesses, while broader sensitivity to partner policies remains open. Correlated histories and missing solver labels also affect the coverage of a training set. We retain these distinctions in evaluation by reporting parent-level coverage, context counts, and reference visitation separately.

7 Research Plan for the Next Semester

The next stage will test whether a curriculum selected by strategic relevance and model difficulty improves coordination. We will first complete the slice specification: the parent-game distribution, model control horizon, continuation policy, and response scoring

objective. The prompt will present the rules, public history, current commitments, and cumulative private answers consistently at every decision. Q/S labels and response rewards will be checked against this common specification.

We will then construct 800 candidate slices from the parent pool and measure D using the base model under fixed prompting and decoding conditions. Selection of the final 100 training slices will combine this measurement with Q/S and structural coverage. The curriculum will include information acquisition, use of existing answers, behavioural inference, and cases where investigation is unnecessary. Splits will be assigned by parent game before model-guided selection to separate related histories across training and evaluation. The new pool has not yet undergone this candidate-generation or D stage.

The main experiment will compare targeted slice training with complete-game self-play and the unchanged model using a common starting checkpoint and equal budgets of generated response tokens. Ablations will examine the contribution of inference and conditional-planning objectives. Training analysis will track distinct sampled actions, nonzero task advantages, and investigation accuracy alongside reward and regret. This directly tests whether the revised curriculum addresses the weaknesses observed in the terminal-slice run.

Evaluation will cover held-out negotiation games and sequential calendar coordination. Negotiation metrics will include action validity, regret, information acquisition, and terminal utility. Scheduling evaluation will measure final retained meetings, complete games, communication, and rescheduling costs. Repeated runs and paired scenarios will quantify variation, and partner-policy tests will examine generalisation beyond same-model teams. Each experiment will preserve its model, data, prompt, and resolved configuration so that comparisons remain reproducible.

8 Conclusion

We investigate strategic game slices as a basis for diagnosing and training coordination under private information. Matched interventions show that correct preference judgments leave substantial planning error, and initial slice-trained checkpoints improve meeting retention in a sequential scheduling task. Training and game-structure analyses further identify sparse information relevance and limited sampled action diversity as distinct obstacles. The three-player studies provide concrete, independently checked examples in which information changes the optimal contract. Building on these results, the next phase will evaluate a curriculum that connects strategic relevance, measured model difficulty, and effective training signal.

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